

SOUTH WEST CAPE, New Zealand

WMO: 939090

Lat: 47.277S

Lon: 167.463E

Elev: 102

StdP: 100.10

Time Zone: 12.00 (NZL)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.0	3.8	-1.7	3.3	5.2	-0.7	3.6	5.6	26.2	10.0	24.5	9.7	10.5	230	1.069

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.8	19.5	15.2	18.1	14.6	17.1	14.0	16.3	18.3	15.4	17.3	14.6	16.4	8.6	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.4	11.1	17.1	14.5	10.4	16.3	13.7	9.9	15.5	46.2	18.1	43.4	17.4	41.3	16.5	20.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.0	23.6	21.5	DB	1.7	23.2	0.8	2.1	1.1	24.7	0.7	25.9	0.2	27.1	-0.3	28.6	(4)
(5)			WB	0.5	17.7	0.8	1.3	-0.1	18.6	-0.6	19.3	-1.0	20.0	-1.6	21.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.3	12.8	13.0	12.4	10.9	9.7	8.3	7.8	8.1	8.9	9.3	10.2	12.0	(6)	
	DBStd	2.71	2.10	2.18	2.15	2.07	2.22	1.85	1.79	1.88	1.97	1.83	2.00	2.18	(7)	
	HDD10.0	344	1	1	3	12	33	56	72	62	44	36	21	5	(8)	
	HDD18.3	2948	173	150	185	222	269	301	327	317	283	281	244	196	(9)	
	CDD10.0	440	87	85	76	40	23	5	3	4	11	13	26	68	(10)	
	CDD18.3	1	1	1	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH23.3	2	1	1	1	0	0	0	0	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	10.7	11.1	10.2	10.2	10.7	11.0	10.8	10.2	9.9	10.9	11.5	11.3	10.4	(14)	
(15) Precipitation	PrecAvg	1338	110	99	115	114	126	115	111	104	107	116	108	108	(15)	
	PrecMax	1703	177	192	297	256	202	191	249	157	175	189	186	180	(16)	
	PrecMin	1162	36	34	48	48	30	45	63	78	45	45	32	57	(17)	
	PrecStd	133	37	44	58	49	44	38	42	19	37	38	37	33	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.2	22.0	20.1	18.1	16.3	13.9	13.1	13.7	15.8	16.4	17.8	20.7	(19)	
		MCWB	16.3	16.7	15.6	13.6	11.5	9.7	9.3	9.2	10.6	11.6	13.6	15.6	(20)	
	2%	DB	19.0	19.2	18.2	16.2	14.9	12.3	12.0	12.3	13.8	14.5	15.9	18.1	(21)	
		MCWB	15.4	15.3	14.8	12.6	11.6	10.1	9.3	8.9	10.1	11.0	12.4	14.1	(22)	
	5%	DB	17.4	17.7	17.0	15.1	13.7	11.6	11.2	11.4	12.6	13.2	14.5	16.7	(23)	
		MCWB	14.4	14.9	14.4	12.6	11.7	9.9	9.2	9.0	9.9	10.5	11.5	13.5	(24)	
	10%	DB	16.2	16.5	15.9	14.1	12.8	11.0	10.5	10.7	11.7	12.1	13.4	15.5	(25)	
		MCWB	13.6	14.1	13.7	12.2	11.3	9.6	8.8	8.8	9.5	10.0	10.8	12.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.5	17.5	17.0	14.7	13.7	11.6	10.9	11.0	11.8	12.7	14.6	16.3	(27)	
		MCDB	19.6	19.9	18.4	16.0	14.5	12.3	11.6	11.9	13.5	14.4	16.7	19.2	(28)	
	2%	WB	16.1	16.2	15.7	13.7	12.8	11.0	10.2	10.1	11.0	11.7	13.1	15.0	(29)	
		MCDB	18.1	18.3	17.4	15.0	13.7	11.8	11.3	11.0	12.6	13.4	14.8	16.9	(30)	
	5%	WB	14.9	15.2	14.7	13.0	12.0	10.3	9.6	9.6	10.3	11.1	12.1	14.1	(31)	
		MCDB	16.8	17.3	16.4	14.5	13.0	11.3	10.7	10.6	11.9	12.7	13.8	16.1	(32)	
	10%	WB	14.0	14.3	13.9	12.2	11.3	9.7	9.0	9.1	9.7	10.3	11.3	13.2	(33)	
		MCDB	15.7	16.2	15.6	13.8	12.5	10.7	10.1	10.3	11.3	11.8	12.9	15.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.9	4.8	4.4	3.7	3.1	3.0	3.0	3.3	3.9	4.3	4.7	5.1	(35)	
		MCDBR	6.9	6.6	6.2	5.2	4.2	3.7	3.7	4.2	5.2	5.8	6.4	6.9	(36)	
	5% WB	MCWBR	4.2	4.3	4.0	3.7	3.0	3.2	2.9	3.0	3.4	4.0	4.3	4.3	(37)	
		MCWBR	6.1	6.0	5.5	4.5	3.5	3.4	3.3	3.4	4.5	5.0	5.7	6.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.345	0.334	0.327	0.314	0.307	0.309	0.310	0.314	0.339	0.335	0.337	0.338	(40)	
		taud	2.541	2.585	2.604	2.619	2.587	2.498	2.535	2.573	2.497	2.530	2.535	2.561	(41)	
	Ebn at Noon		972	952	908	839	753	691	735	824	876	941	974	987	(42)	
		Edh at Noon	104	94	83	68	56	54	58	69	90	98	104	104	(43)	
(44) All-Sky Solar Radiation	RadAvg		5.53	4.62	3.30	2.08	1.18	0.88	1.07	1.79	2.86	4.23	5.24	5.84	(44)	
	RadStd		0.40	0.26	0.21	0.14	0.07	0.07	0.07	0.11	0.14	0.26	0.38	0.31	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	+0.60	N/A	N/A	N/A	-53	-132	+76	N/A

Nomenclature: See separate page